CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

Project Report

Project Title:
Credit Card Fraud Detection Using Machine Learning

Submitted By:
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Technology Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn

Machine Learning Algorithm:

- Logistic Regression

Dataset:
Credit Card Fraud Detection Dataset

Academic Year:
2025-26

Certificate

This is to certify that the project entitled **"Credit Card Fraud Detection Using Machine Learning"** has been successfully completed as part of the internship/project work requirement. The project demonstrates the application of Machine Learning techniques for detecting fraudulent credit card transactions.

Acknowledgement

I would like to express my sincere gratitude to my mentor, faculty members, and all those who guided me throughout the development of this project. Their valuable suggestions and support helped me complete this work successfully.

Abstract

Credit card fraud has become a major challenge in modern digital payment systems. The increasing volume of online transactions creates opportunities for fraudulent activities, causing financial losses for banks and customers.

This project develops a Machine Learning-based fraud detection system using Logistic Regression. The model is trained on a publicly available credit card transaction dataset containing both genuine and fraudulent transactions. Various data analysis techniques, visualization methods, and performance evaluation metrics are used to assess the model's effectiveness.

The developed model achieved an accuracy of approximately **99.93%** and an ROC-AUC score of **0.95**, demonstrating strong performance in distinguishing fraudulent transactions from legitimate ones.

1. Introduction

With the rapid growth of online banking and electronic payment systems, credit card fraud has become a significant concern. Financial institutions require intelligent systems capable of identifying suspicious transactions in real time.

Machine Learning provides powerful techniques for analyzing transaction patterns and automatically detecting fraudulent behavior.

This project focuses on building a fraud detection system using Logistic Regression.

2. Problem Statement

Traditional rule-based fraud detection systems often fail to identify new fraud patterns and generate many false alarms.

The objective of this project is:

- To identify fraudulent credit card transactions.
- To minimize false detections.
- To improve transaction security using Machine Learning.

3. Objectives

The main objectives are:

1. Analyze credit card transaction data.
2. Identify fraudulent and genuine transactions.
3. Train a Machine Learning model.

4. Evaluate model performance.
5. Predict fraud probability for new transactions.
6. Reduce financial losses due to fraud.

4. Dataset Description

Dataset Used:
Credit Card Fraud Detection Dataset

Dataset Statistics

Parameter	**Value**
Total Transactions	284,807
Genuine Transactions	284,315
Fraud Transactions	492
Features	30

Target Variable	**Class**
Class 0	Genuine
Class 1	Fraud

Observations

The dataset is highly imbalanced:

- Genuine Transactions = 99.83%
- Fraud Transactions = 0.17%

This imbalance makes fraud detection a challenging classification problem.

5. Methodology

Step 1: Data Collection

The credit card transaction dataset was collected and loaded into Pandas DataFrame.

Step 2: Data Exploration

Performed:

- Dataset shape analysis
- Missing value analysis
- Statistical summary
- Class distribution analysis

Step 3: Data Visualization

Three visualizations were created:

1. Transaction Distribution Graph
2. Confusion Matrix
3. ROC Curve

Step 4: Data Splitting

Dataset divided into:

- Training Data = 80%
- Testing Data = 20%

Result:

Dataset	**Shape**
Training	(227845,30)
Testing	(56962,30)

Step 5: Model Training

Algorithm Used:

Logistic Regression

Reasons:

- Fast training
- Easy implementation
- Good binary classification performance

- Suitable for fraud detection

Step 6: Model Evaluation

Metrics Used:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve
- AUC Score

6. Results and Discussion

6.1 Transaction Distribution Analysis

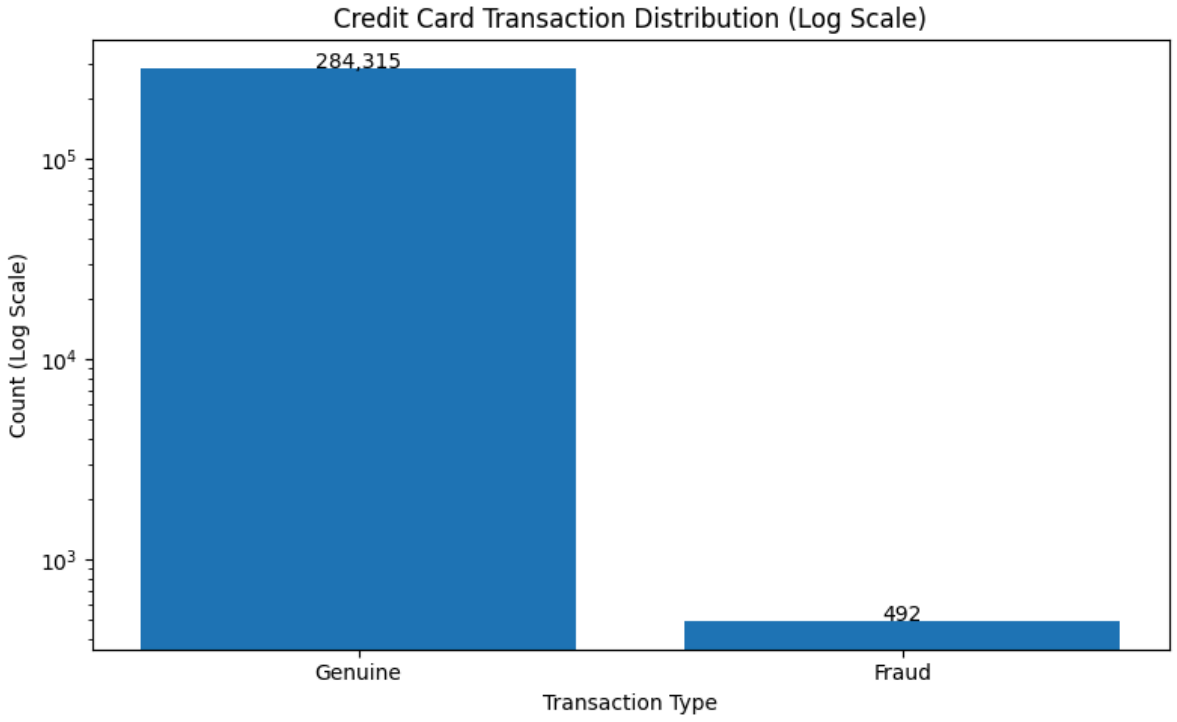


Figure 1: Credit Card Transaction Distribution

Observation:

- Genuine transactions = 284,315
- Fraud transactions = 492

The graph clearly shows a severe class imbalance.

Interpretation:

Most transactions are legitimate, while only a very small fraction are fraudulent.

6.2 Confusion Matrix Analysis

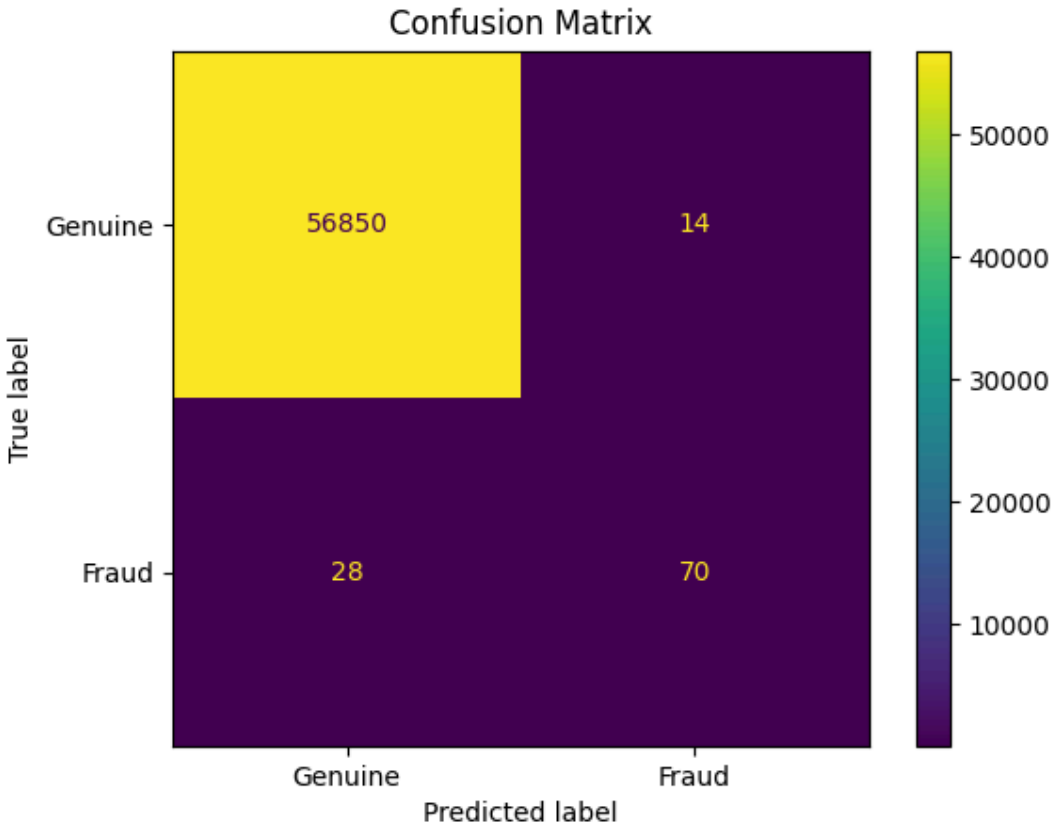


Figure 2: Confusion Matrix

	Predicted Genuine	**Predicted Fraud**
Actual Genuine	56850	14
Actual Fraud	28	70

Interpretation

- **True Negatives (TN)**
  - 56850 transactions correctly identified as genuine.
- **True Positives (TP)**
  - 70 fraudulent transactions correctly identified.
- **False Positives (FP)**
  - 14 genuine transactions incorrectly classified as fraud.

- **False Negatives (FN)**
    - 28 fraudulent transactions missed by the model.

Observation

The model successfully detected most fraudulent transactions while maintaining a very low false alarm rate.

6.3 ROC Curve Analysis

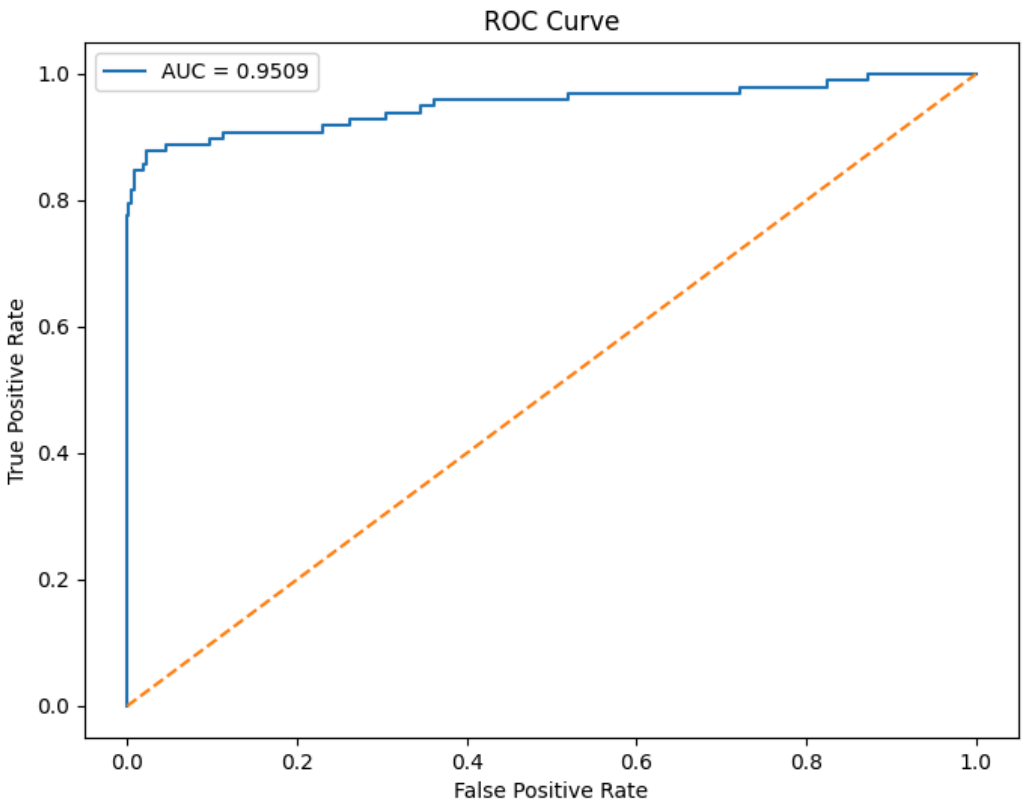


Figure 3: ROC Curve

ROC-AUC Score:

AUC = 0.9509

Interpretation

An AUC value close to 1 indicates excellent classification performance.

The model achieves:

- High True Positive Rate

- Low False Positive Rate

Thus, the model effectively separates fraud and genuine transactions.

7. Classification Report

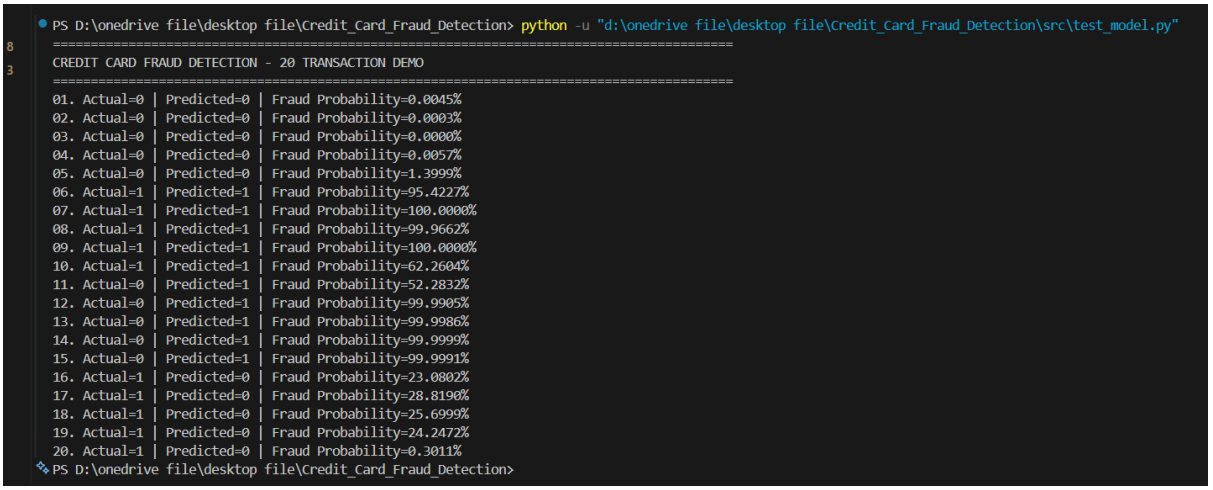
Metric	Genuine (0)	Fraud (1)
Precision	1.00	0.83
Recall	1.00	0.71
F1-Score	1.00	0.77

Overall Accuracy:
99.93%

Analysis

- Precision for fraud detection = 83%
- Recall for fraud detection = 71%
- Strong overall model performance

8. Transaction Testing Results



A demonstration was conducted using 20 transactions representing all four possible prediction scenarios.

True Negative Examples

Actual	Predicted
0	0
0	0
0	0
0	0
0	0

These are genuine transactions correctly classified.

True Positive Examples

Actual	Predicted
1	1
1	1
1	1
1	1
1	1

These are fraud transactions correctly detected.

False Positive Examples

Actual	Predicted
0	1
0	1
0	1
0	1
0	1

These are genuine transactions incorrectly flagged as fraud.

False Negative Examples

Actual	Predicted
1	0
1	0
1	0
1	0
1	0

These are fraudulent transactions missed by the model.

9. Advantages

1. High prediction accuracy.
2. Fast processing.
3. Easy deployment.
4. Low computational cost.
5. Effective fraud identification.
6. Suitable for real-time systems.

10. Limitations

1. Dataset is highly imbalanced.
2. Some fraud transactions remain undetected.
3. Logistic Regression may not capture highly complex fraud patterns.
4. Performance may decrease with evolving fraud strategies.

11. Future Scope

1. Apply SMOTE for class balancing.
2. Use Random Forest.
3. Use XGBoost.
4. Implement Deep Learning models.
5. Real-time fraud monitoring system.
6. Deploy as a web application.

12. Conclusion

This project successfully developed a Machine Learning-based Credit Card Fraud Detection System using Logistic Regression.

The model achieved:

- Accuracy = 99.93%
- ROC-AUC = 0.9509
- Strong fraud detection capability

The experimental results demonstrate that Machine Learning can significantly improve fraud detection performance while reducing false alarms. The system can be further enhanced using advanced ensemble and deep learning approaches.

13. References

1. Scikit-Learn Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Credit Card Fraud Detection Dataset (Kaggle)
6. Machine Learning Research Papers on Fraud Detection